

1: Differences Between Ethernet Cables and Fiber Optic Cables

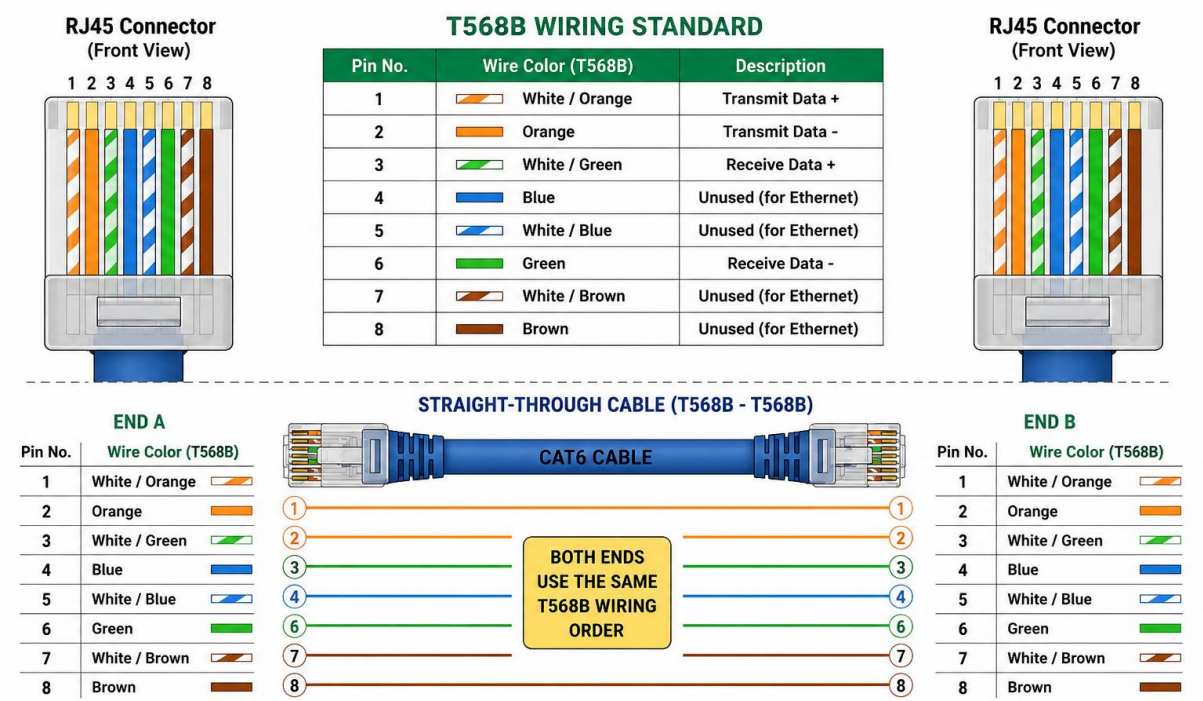
Feature	Ethernet Cable	Fiber Optic Cable
Transmission Medium	Electrical signals through copper wires	Light signals through glass/plastic fibers
Speed & Distance	Up to 1–10 Gbps, maximum 100 meters	10 Gbps to 100+ Gbps over several kilometers
Cost	Lower cost, easy to install	Higher cost, requires specialized tools

Examples:

- Ethernet Cable: Connecting a home Wi-Fi router to a desktop PC.
- Fiber Optic Cable: Connecting servers inside a data center or a college campus backbone network.

2: Ethernet Cable

STRAIGHT-THROUGH ETHERNET CABLE (T568B STANDARD)



3: Steps to Attach an RJ45 Connector to a CAT6 Cable

1. Cut the CAT6 cable to the required length using a cable cutter.
2. Strip about 2–3 cm of the outer jacket using a wire stripper.
3. Untwist the wire pairs and straighten all eight wires carefully.
4. Arrange the wires in the T568B order:
 - White/Orange
 - Orange
 - White/Green
 - Blue
 - White/Blue
 - Green
 - White/Brown
 - Brown
5. Trim the wires evenly so they are the same length.
6. Insert the wires into the RJ45 connector, ensuring each wire reaches the end of the connector and remains in the correct order.
7. Place the RJ45 connector into the RJ45 crimping tool.
8. Squeeze the crimper firmly to secure the connector to the cable.
9. Test the completed cable with a cable tester to verify all eight pins are connected correctly.

4: Gaming Cafe Network Setup (10 PCs)

For a gaming café with 10 LAN gaming PCs, I would choose CAT6 Ethernet cables because they provide high speed (up to 1–10 Gbps), low latency, and are affordable for short distances inside a room. I would use an RJ45 crimper to attach RJ45 connectors to the CAT6 cables. Fiber optic cables would be unnecessary because the distances are short and fiber installation is much more expensive. Ethernet offers the best balance of performance, reliability, and budget for this setup.